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COURSE : ITA 0612 - MACHINE LEARNING.

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01. Given:-

↳ Box A = 6 Red, 4 Blue, 2 Green → Total = 12.

↳ Box B = 5 Red, # Blue, 3 Green → Total = 15

↳ Box C = 8 Red, 2 Blue, 5 Green → Total = 15

$\Rightarrow P(A) = \frac{1}{4}$ $P(B) = \frac{1}{2}$ $P(C) = \frac{1}{4}$

Probability of blue:

- $P(Blue/A) = 1/3$
- $P(Blue/B) = \#/15$
- $P(Blue/C) = 2/15$

Using Bayes Thm.

$$P(C/Blue) = \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{\#}{15} + \frac{1}{4} \times \frac{2}{15}}$$

$$= \frac{2}{21}$$

$P(C/Blue) = 0.0952$

02] Given:

$P(A) = \frac{1}{4}$ $P(B) = \frac{1}{2}$ $P(C) = \frac{1}{4}$.

Blue Probabilities :-

- $P(\text{Blue}/A) = 3/10$
- $P(\text{Blue}/B) = 2/15$
- $P(\text{Blue}/C) = 2/15$

$$\Rightarrow P(C/\text{Blue}) = \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15}}$$

$$= \frac{4}{37}$$

$P(C/\text{Blue}) = 0.1081$ Ans.

03. Given:

- $TP = 210$
- $TN = 190$
- $FP = 30$

$\therefore FN = 360 - 210 = 150.$

$$\Rightarrow \text{Accuracy} = \frac{TP + NN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720}$$

$$= 55.56\%$$

$$\Rightarrow \text{Specificity} = \frac{TN}{TN + FP} = \frac{190}{190 + 30}$$

$$= 86.36\%.$$

04. Given :-

- TP = 119
- FN = 6
- TN = 120
- FP = 5

↳ Accuracy = $\frac{TP+TN}{Tot} = \frac{119+120}{250} = 95.6\%$

↳ Precision = $\frac{TP}{TP+FP}$

$= \frac{119}{119+5} = 95.9\%$

↳ Sensitivity = $\frac{TP}{TP+FN} = \frac{120}{120+5}$

$= 96\%$.

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05. Given:-

TP = 138

FN = 112

TN = 135

FP = 15.

Accuracy :-

$$= \frac{TP + TN}{Tot} = \frac{138 + 135}{300}$$

$$= 91\%$$

Precision. $= \frac{TP}{TP + FP}$

$$= \frac{138}{138 + 15}$$

$$= 90.2\%$$

Specificity : $\frac{TP}{TP + FN}$

$$= \frac{135}{138 + 12}$$

$$= 92\%$$

F1 - Score.

$$= \frac{2(138)}{2(138) + 15 + 12}$$

$$= 91.09\%.$$

Ans